Exercises

Exercise 5.1: Epidemiology

Imagine that you are an epidemiologist and you are determining people's cause of death.

In this simplified world, there are two main diseases, cancer and the common cold.

- People rarely have cancer, $\Pr(\mathsf{Cancer}) = 0.00001$.
- People are much more likely to have a common cold: $\Pr(\mathsf{Cold}) = 0.2$.
- When people do have cancer, it is often fatal: $\Pr(\mathsf{Death} \mid \mathsf{Cancer}, \neg\mathsf{Cold}) = 0.9$.
- Having a cold is rarely fatal: $\Pr(\mathsf{Death} \mid \mathsf{Cold}, \neg\mathsf{Cancer}) = 0.00006$
- But having cancer and a cold is slightly more deadly than having cancer alone: $\Pr(\mathsf{Death} \mid \mathsf{Cold}, \mathsf{Cancer}) = 0.91$.
- Very rarely, people also die of other causes: $\Pr(\mathsf{Death} \mid \neg\mathsf{Cold}, \neg\mathsf{Cancer}) = 0.000000001$.

Write this model in WebPPL and use `Infer` to answer the following questions (Be sure to include your code in your answer):

Question 5.1.1

Compute $\Pr(\mathsf{Cancer} \mid \mathsf{Death}, \mathsf{Cold})$ and $\Pr(\mathsf{Cancer} \mid \mathsf{Death}, \neg\mathsf{Cold})$.

How do these probabilities compare to $\Pr(\mathsf{Cancer} \mid \mathsf{Death})$ and $\Pr(\mathsf{Cancer})$?

Using these probabilities, give an example of explaining away.

```
display("Prior:")
viz.table(
  Infer(
    {method: 'enumerate'},
    function() {
      ...
    }
  )
)
```

```
display("Death:")
viz.table(
  Infer(
    {method: 'enumerate'},
    function() {
      ...
    }
  )
)
```

```
display("Death and Cold:")
viz.table(
  Infer(
    {method: 'enumerate'},
    function() {
      ...
    }
  )
)
```

```
display("Death and No Cold:")
viz.table(
  Infer(
    {method: 'enumerate'},
    function() {
      ...
    }
  )
)
```

Question 5.1.2

Compute $\Pr(\mathsf{Cold} \mid \mathsf{Death}, \mathsf{Cancer})$ and $\Pr(\mathsf{Cold} \mid \mathsf{Death}, \neg\mathsf{Cancer})$.

How do these probabilities compare to $\Pr(\mathsf{Cold} \mid \mathsf{Death})$ and $\Pr(\mathsf{Cold})$?

Using these probabilities, give an example of explaining away.

```
display("Prior:")
viz.table(
  Infer(
    {method: 'enumerate'},
    function() {
      ...
    }
  )
)
```

```
display("Death:")
viz.table(
  Infer(
    {method: 'enumerate'},
```

```
    function() {
      ...
    }
  )
)
```

```
display("Death and Cancer:")
viz.table(
  Infer(
    {method: 'enumerate'},
    function() {
      ...
    }
  )
)
```

```
display("Death and No Cancer:")
viz.table(
  Infer(
    {method: 'enumerate'},
    function() {
      ...
    }
  )
)
```

Exercise 5.2: Are They Independent?

For each question, determine whether the relevant variables are independent in the given Bayes Net. Here, grey nodes represent *observed* variables while white nodes represent *unobserved* variables.

Question 5.2.1

Are A and B independent?

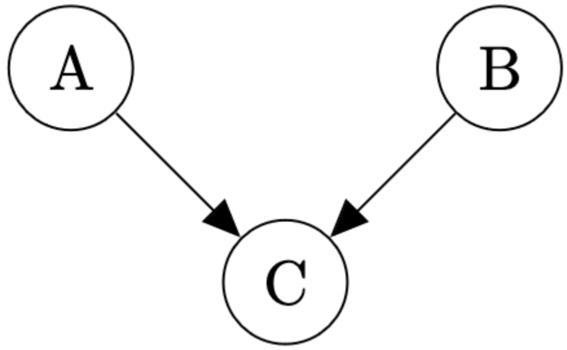

Figure 1: Bayes Net for Question 5.2.1

Question 5.2.2

Are A and B independent?

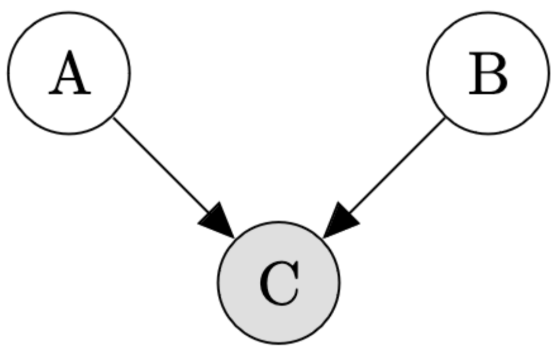

Figure 2: Bayes Net for Question 5.2.2

Question 5.2.3

Are A and C independent?

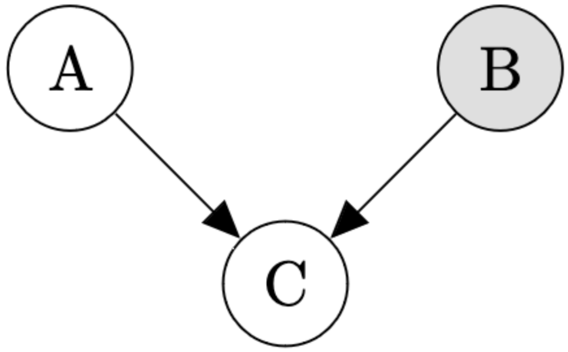

Figure 3: Bayes Net for Question 5.2.3

Question 5.2.4

Are B and C independent?

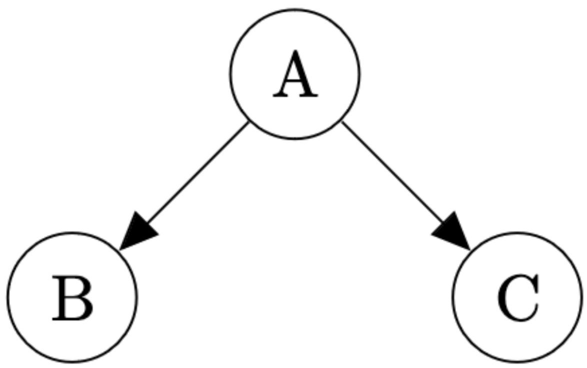

Figure 4: Bayes Net for Question 5.2.4

Question 5.2.5

Are B and C independent?

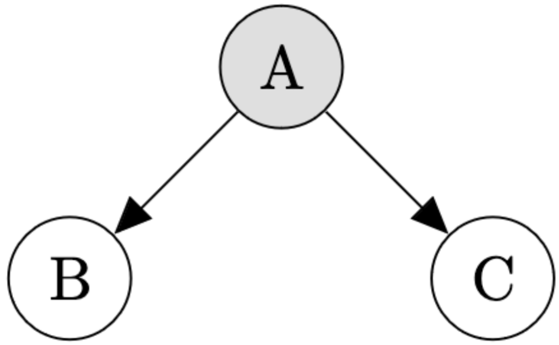

Figure 5: Bayes Net for Question 5.2.5

Question 5.2.6

Are A and B independent?

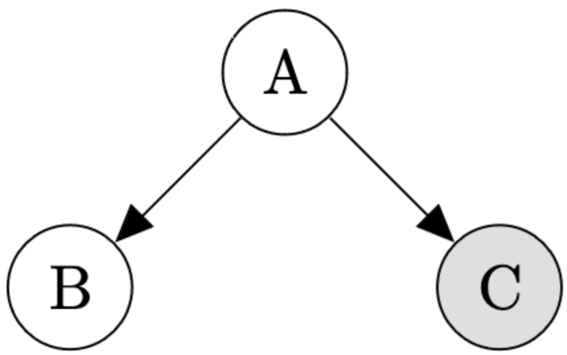

Figure 6: Bayes Net for Question 5.2.6

<script> document.addEventListener("DOMContentLoaded", () => document.querySelectorAll('pre').forEach((elt) => editor.setup(elt, language:'webppl'))); </script>